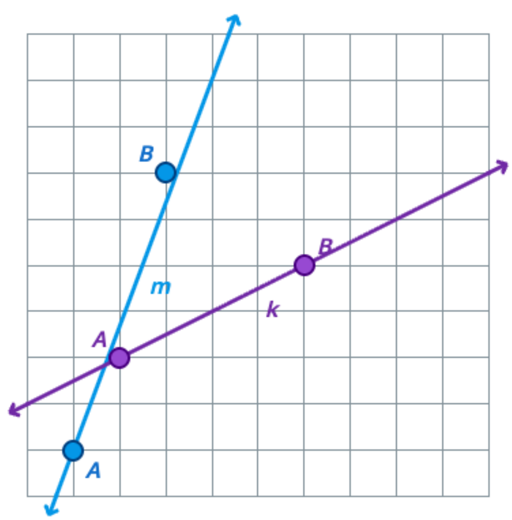
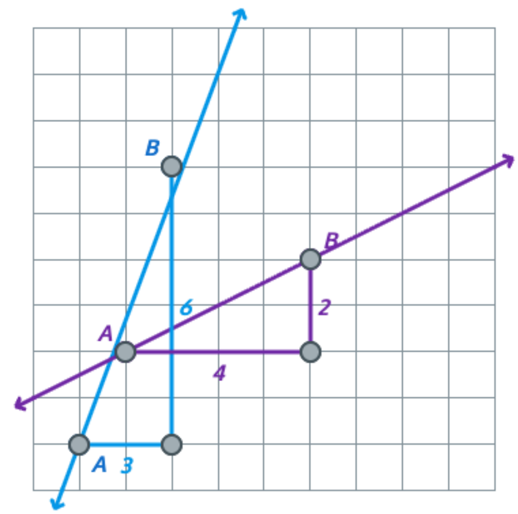
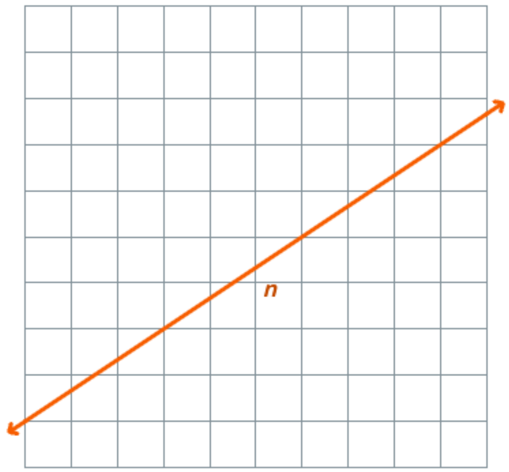
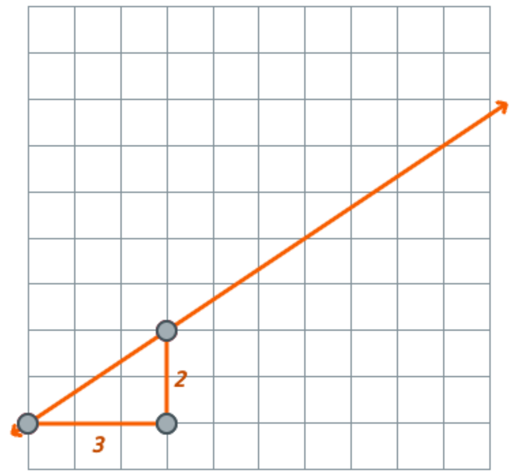
Lesson Objective

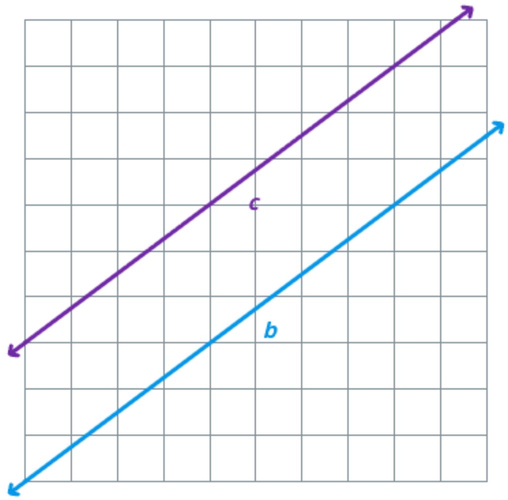
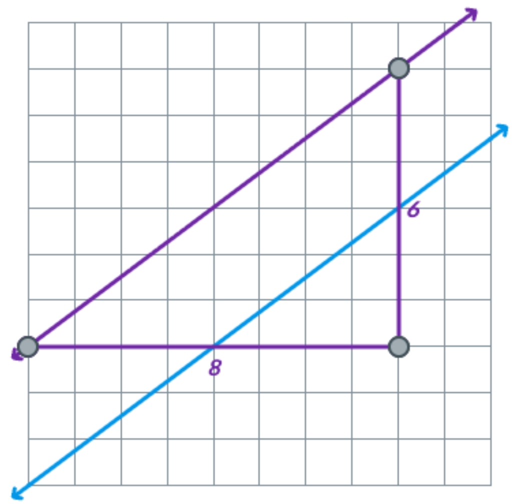
Find the slope of a line on a grid using properties of slope triangles.

Materials: Straightedges

1. Compare the steepness of two lines on a grid and find the slope of each using a slope triangle. Which line is steeper, line m or line k? Then find the slope of each line by drawing a slope triangle between points A and B on each line.  *Answer: Line m is steeper; slope of line m = 6/3 = 2; slope of line k = 2/4 = 1/2.*	Ask students which line looks steeper and how they can tell. Tell students that the slope of a line is a number that describes its steepness, and that we can find it by drawing a slope triangle between two points and dividing the vertical distance by the horizontal distance. Model drawing a slope triangle on line m from point A to point B, with a horizontal leg and a vertical leg along the grid lines. Have students count the horizontal distance (3 units) and the vertical distance (6 units), then record the slope as 6/3, or 2. Guide students through drawing a slope triangle on line k from point A to point B. Have them count the horizontal distance (4 units) and the vertical distance (2 units), then record the slope as 2/4, or 1/2. Point out that line m has the larger slope and is the steeper line. Tell students that a larger slope

	means a steeper line because the vertical change is greater for each unit of horizontal change. 
2. Find the slope of a line on a grid by drawing a slope triangle between two points. Find the slope of line n. Choose two points where the line crosses the intersection of grid lines, then draw a slope triangle to help you.	Have students pick two points on line n where the line crosses the intersection of grid lines. Ask why that choice makes it easier to count the distances. Ask students to draw a slope triangle between their two points, with one leg horizontal and one leg vertical. Have them count the horizontal distance and the vertical distance and record the slope as vertical distance divided by horizontal distance. Check that they get a slope of 2/3. Ask students what would happen if they chose a different pair of lattice points on line n. Point out that the

 *Answer: Slope of line n = 2/3 (e.g., from (0, 1) to (3, 3), vertical distance 2 and horizontal distance 3).*	slope triangles between any two points on the same line are similar, so the ratio of vertical to horizontal is always the same. 
3. Find the slope of a line parallel to a given line by drawing and comparing slope triangles. Lines b and c are parallel. The slope of line b is 3/4. Find the slope of line c by drawing a slope triangle on line c. What do you notice?	Remind students that the slope of line b is 3/4, which means for every 4 units of horizontal distance, the vertical distance changes by 3 units. Have students choose two lattice points on line c and draw a slope triangle between them. Have them count the vertical and horizontal distances and record the slope. Ask students to compare the slope of line c to the slope of line b. Point out that the two slopes are equal. Tell students that parallel lines always have the same slope because their slope

 *Answer: Slope of line c = 6/8 = 3/4. The slopes are the same; parallel lines have equal slopes.*	triangles have the same shape, so the ratio of vertical to horizontal is identical. 

Monitor For	**Instructional Moves**
• Can students name which line is steeper in Problem 1 and connect that to having the larger slope? • Do students draw the slope triangle with one horizontal leg and one vertical leg, rather than drawing along the line itself? • Do students record slope as vertical distance divided by horizontal distance, not the other way around?	• If a student counts along the slanted line instead of along the grid, redirect by tracing the horizontal leg with a finger and asking, "How many units across?" then tracing the vertical leg and asking, "How many units up?" • Reinforce the slope ratio by writing "vertical / horizontal" next to each slope as students record it (e.g., 6/3 = 2 for line m). • If a student picks two points that are not at grid intersections in Problem 2, prompt them to

<table>
<tr><td>• In Problem 2, do students choose two points where the line crosses the intersection of grid lines so they can count whole units?</td><td>choose points where the line meets a corner of the grid so the side lengths are whole numbers.

• Make the parallel-lines idea explicit in Problem 3 by asking, "What is the slope of line b? What is the slope of line c? What do you notice about the two slopes?"</td></tr>
<tr><td colspan="2">Reflection Questions

• How did you use a slope triangle to find the slope of line m? Answer: I drew a triangle from point A to point B with a horizontal leg and a vertical leg. The horizontal distance was 3 and the vertical distance was 6, so the slope is 6/3, or 2.

• Line m had a slope of 2 and line k had a slope of 1/2. How did the slopes match what you saw about steepness? Answer: Line m was steeper, and it had the larger slope. A larger slope means the line goes up more for each unit across.

• In Problem 2, would you get the same slope if you picked a different pair of points on line n? Why? Answer: Yes. The slope triangles between any two points on the same line are similar, so the ratio of vertical to horizontal is the same.

• What did you notice about the slopes of lines b and c? Why does that make sense? Answer: The slopes were equal. Parallel lines never meet, so they have the same steepness, which means the same slope.</td></tr>
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